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RADIATION RESEARCH 117, 379–394 (1989)

ESR Characteristics of One-Electron Reduced Thymine in Monomer, Oligomer, and Polymer Derivatives

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ESR spectra have been recorded of one-electron reduced thymine ($T^{\bullet-}$) trapped in a variety of thymine derivatives and matrices. Most of the data were taken at *Q*-band microwave frequencies on 12 *M* LiCl glasses containing oligo(T) varying from 2 to 18 nucleotides in length. A nonlinear least-squares simulation program was used to simulate the ESR spectra, resulting in a parameter set that can be used to generate $T^{\bullet-}$ powder spectra at *X*- and *Q*-band frequencies. Although it is not possible to identify a unique set of *g* and hyperfine tensors, it is possible to establish boundaries for these interactions and propose a parameter set that is likely and reasonable. As the trapping molecule is varied from TdR to oligo(T) to poly(T), there are small changes in the interaction parameters. The changes are not large enough to account for the differences between hfc values determined from DNA fiber data versus hyperfine coupling values measured from monomer systems. The analysis also indicates that, in these systems, $T^{\bullet-}$ does not protonate between 4 and 77 K at neutral pH. © 1989 Academic Press, Inc.

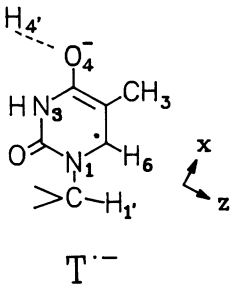
INTRODUCTION

Two or more free radical products are trapped in duplex DNA when it is exposed to the direct effects of ionizing radiation at temperatures of 77 K or below. One of these radicals, the thymine radical anion (Structure I), has been identified with a high degree of certainty. The most persuasive evidence comes from measurements that demonstrate the conversion of an ESR doublet into the readily identifiable eight-line spectrum of the 5,6-dihydrothymine-5-yl spectrum (*1*). This conversion, due to the protonation of $T^{\bullet-}$ at C6, in conjunction with the doublet structure of the ESR spectrum (*2–4*) leaves little question that the $T^{\bullet-}$ radical is trapped in DNA. There are questions, however, as to what fraction of the free radical population is assignable to $T^{\bullet-}$, as to whether $T^{\bullet-}$ protonates at OH prior to protonation at C6, and as to the identity of the other free radicals trapped in DNA (*5*). To answer these questions using ESR spectroscopy, it is important to determine what the characteristics of the $T^{\bullet-}$ radical are when it is trapped in the environment of the DNA duplex.

Much of the pioneering work on radical ions produced in the nucleic bases was done by Sevilla and co-workers (*2–4*). They studied electron attachment in glasses, such as 8 *M* NaOD and 12 *M* LiCl, using photooxidation of $K_4Fe(CN)_6$ to generate electrons. Others have employed ionizing radiation to generate radical anions of the bases in various glasses (*6–8*). The ESR studies of $T^{\bullet-}$ in these systems indicate that the major hyperfine coupling, A(H6), varies with conditions. A proton on O4 (see

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STRUCTURE I

Structure I) increases A_{iso}(H6), perhaps by as much as ~16 MHz. A ribose substituent at N1 decreases A_{iso}(H6) by ~6 MHz. The magnitude of these changes is large enough to affect the analysis of $T^{\bullet-}$ trapped in DNA.

In addition to the glassy systems, ESR measurements on $T^{\bullet-}$ have been made in single crystals, aqueous solutions, and DNA fibers. Table I summarizes these results. A precise measurement of the A(H6) tensor was obtained via an electron nuclear double resonance (ENDOR) study of thymidine crystals (*9*). The isotropic hyperfine couplings (hfc's) due to H6, N1, N3, CH_3, H1, and H3 in thymine were measured using aqueous radiolysis ESR (*10*). The tensors for H6, N1, CH_3, H1', and *g* were estimated from DNA fiber data (*11, 12*). The fiber system has the advantage that the $T^{\bullet-}$ radicals are partially aligned and the disadvantage that the $T^{\bullet-}$ spectrum is superimposed on the spectrum of one or more other radicals. A best fit of the fiber data is obtained using a free radical population consisting of 50% thymine radical anion and 50% guanine radical cation ($G^{\bullet+}$). There is some debate over this $T^{\bullet-}/G^{\bullet+}$ hypothesis (*5, 13*). One can see from Table I that the ESR interactions determined from the DNA fiber system do not agree well with other systems. The difference in $A_{iso}(CH_3)$ is particularly notable.

It would be advantageous in the analysis of DNA if the ESR spectrum of $T^{\bullet-}$ was known more precisely. It is clear that the spectrum of $T^{\bullet-}$ varies with chemical environment and substituents. We set out, therefore, to determine a set of parameters that will generate the spectrum of $T^{\bullet-}$ trapped in duplex DNA. The strategy is first to determine the ESR hyperfine and *g* interactions in $T^{\bullet-}$, or at least better define the boundaries of the interactions, and then to observe how these interactions are influenced by environment and conformation. The method is to study $T^{\bullet-}$ generated in 12 *M* LiCl glasses containing various thymine derivatives. We will return to the results given in Table I in the Discussion.

METHODS

Samples. Thymidine, 1-methyl-thymine, $d(pT)_2$, $d(pT)_3$, $d(pT)_4$, $d(pT)_{10}$, $d(pT)_{12-18}$, and poly(T) were obtained from Sigma Chemical Co. The 1,3-dimethylthymine was graciously provided by S. Steenken, Max-Planck Institut für Strahlenchemie, Mülheim/Ruhr, West Germany. High purity LiCl, LiBr, and HCl were purchased from Alpha Products. These materials were used without further purification. The water was distilled, deionized, and filtered. Although the stock 12 *M* LiCl solution was degassed by freeze–thaw, this had no observable effect on the ESR spectra. The glasses were made by adding 10 μl of 12 *M* LiCl to

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TABLE I

Measurements of g and Hyperfine Coupling Interactions in $T^{\cdot-}$ under Various Conditions[a]

Row	*Matrix*	*g*	*H6*	*N1*	*N3*	CH_3	*H1'*	*H3*	W^b	*Ref.*
1	TdR									
	12 *M* LiCl, 77 K		(39.2)							(*4*)
2	TdR		13.0, 29.9, 55.5							
	crystal 4 K		(32.8)							(*9*)
3	T									
	H_2O RT	—	(33.04)	(3.78)	(2.52)	(2.52)	—	(0.56)2×	—	(*10*)
4	U									
	H_2O RT	(2.0030)	(35.42)	(4.20)	(2.52)	(2.52)[c]	—	(2.52)	—	(*10*)
5	UMP									
	H_2O RT	—	(34.72)	(2.94)	(0.70)	(5.32)[c]	—	(0.70)	—	(*10*)
6	DNA	2.0040, 2.0030, 2.0030	19.6, 40.3, 53.2	1.3, 8.4, 1.3	—	—	—	—	12.3, 22.4, 12.3	
	fiber 77 K	(2.0033)	(37.7)	(3.7)	(0.0)	(10.1)	(0.0)	(0.0)	(15.7)	(*11*)
7	DNA	2.0039, 2.0021, 2.0034	19.6, 35.8, 58.8	0.0, 11.8, 0.0	—	—	−4.5, 9.0, −4.5	—		
	fiber 77 K	(2.0033)	(38.1)	(3.9)	(0.0)	(10.9)	(0.0)	(0.0)	(10.6)	(*12*)
	Oligo(T)	2.0037, 2.0020, 2.0030	16.0, 35.7, 57.9	1.9, 4.5, 1.0	0.6, 1.3, 0.3				0.8, 10.1, −6.1	
8[d]	12 *M* LiCl, 77 K	(2.0030)	(36.5)	(2.5)	(0.7)	(3.7)	(2.0)	(4.0)	(1.6)	e

[a] Hyperfine couplings are in MHz. Principal values are separated by commas. Isotropic values are enclosed in parentheses.
[b] Linewidth tensor.
[c] Hyperfine couplings due to H5.
[d] The principal values are not coaxial. The small hfc's are estimates based on indirect arguments (see text).
[e] This work.

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premeasured quantities of solute. For the oligomers, 5 units (OD at 260 nm) in 10 μl was typical, giving a concentration of $\sim$0.3 m*M*. The concentration of TdR, however, was varied from 2 to 100 m*M*. An 8-μl aliquot was injected into a 1.0-mm Spectrosil quartz tube having a 0.1-mm wall (Friedrich and Dimmock). The small amount of quartz relative to sample rendered the quartz background at 4 K negligible.

The preparation of poly(T) samples differed because of difficulties in injecting the sample into a 1.0-mm tube without loss of polymer. Instead, poly(T) was first dissolved in H_2O, then injected into the 1.0-mm tube, then mixed with 24 *M* LiCl in a 1:1 ratio. The resulting sample was of a gel-like consistency.

ESR. Most of the data were collected at *Q*-band microwave frequencies using a Varian E-12 spectrometer. At *Q* band, X irradiation was done using a Machlett OEG 60 X-ray tube operating at 50 kV and 20 mA, giving 6 μGy/h at the sample position. Irradiation time was 15 min. The Dewar type cryostat used for irradiation and ESR measurement is described elsewhere (*14*). The irradiation temperature was $\sim$2.5 K and ESR data were collected at 4 K. To avoid rapid passage effects, the sweep time was 30 min and the modulation frequency was 1.0 kHz. The $\sim$20 mW microwave power source was attenuated $\sim$50 db. Because these settings give a very weak AFC signal, it was necessary to check for distortion of the spectra. This was done by monitoring a reference, a ruby crystal mounted on the side wall of the cavity, with each sweep. If the ruby signal deviated from proper symmetry, the spectrum was retaken.

Some data were collected at *X*-band microwave frequencies using an IBM/Bruker 200 DRC. The radiation source and total dose were the same but the dose rate was $\sim$30 μGy/h. The cryostat was a Joule–Thompson-type refrigerator that provides an irradiation temperature of $\sim$11 K and measurement temperature of 6 K. As with the *Q* band, low power and low frequency modulation were employed to avoid rapid passage effects.

Simulations. ESR spectra were simulated using a nonlinear least-squares procedure described elsewhere (*15*). This procedure fits between one and six ESR spectra simultaneously using a spin hamiltonian that contains a *g*-tensor and up to seven hyperfine tensors. The tensors need not be coaxial. The linewidth is treated as a tensor and is a peak-to-peak value of the first derivative of a Gaussian lineshape. The simulations reported here utilized 890 orientations of the radical relative to the external magnetic field (B_0). Background spectra were obtained using the sample preparations described above without the thymine derivative. Background subtraction and baseline corrections were made as part of the nonlinear least-squares procedure.

It is useful to define a quantity, *R*, that provides a measure of the goodness-of-fit while remaining insensitive to spectral structure and intensity. We adopt for *R* the definition used for the residual index in X-ray structure

$$R = \sum_i |Y_{oi} - Y_{ci}| / \sum_i |Y_{oi}|, \tag{1}$$

where Y_{oi} are the data points making up the ESR spectrum and Y_{ci} are the calculated values. Typically, when $R < 0.05$ the fit is very good. A fit obtained with an $R = 0.01$ is shown in Fig. 1.

When fitting the $T^{\bullet-}$ spectrum, indeed most powder spectra of organic radicals, there are numerous parameter sets that will give low *R* values. The problem of multiple minima is a common one; the spectra have insufficient structure to readily distinguish a global minimum in the least-squares differences. This problem of multiple minima can be reduced, but not eliminated, by several features of the program. One is to fit more than one spectrum at a time, particularly spectra taken at different microwave frequencies. This reduces the number and range of local minima by introducing additional data that vary in a way predicted by the spin hamiltonian. In the case of $T^{\bullet-}$ the **g** and the **A**(H6) tensors both give rise to significant variations in the *Q*-band spectrum versus the *X*-band spectrum. Another important feature is that each parameter being varied under the least-squares procedure can be confined to a designated window. This can be used to search only that portion of parameter space that is considered realistic.

RESULTS AND ANALYSES

Spectra of $T^{\bullet-}$. The ESR spectrum of $T^{\bullet-}$ varies with matrix, substituents, and conformation. Examples are shown in Fig. 2. In Fig. 2a, the effect of changing from H_2O to D_2O gives a slight narrowing; in Fig. 2b, changing from 12 *M* LiCl to 12 *M* LiBr gives broadening; in Fig. 2c, changing to strong acid (3 *M* HCl + 5 *M* LiCl) gives

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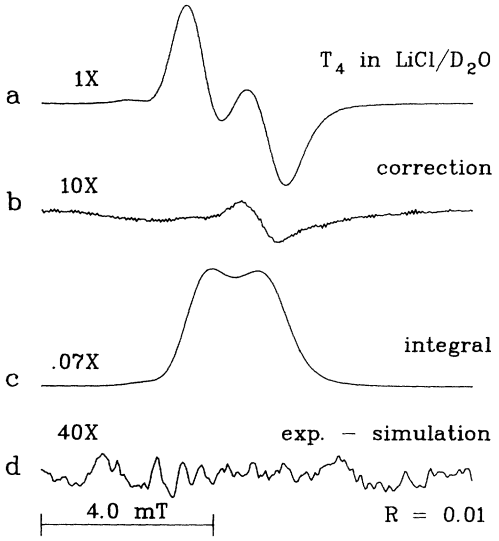


FIG. 1. Experimental and simulated ESR spectrum of $T^{\bullet-}$. a, ESR spectrum taken at 4 K and 35 GHz; b, correction due to subtraction of background and fourth-order polynomial; c, integral of the corrected spectrum; d, difference between the experimental spectrum and a simulation giving $R = 0.01$. Each trace was multiplied by the factor given on the left.

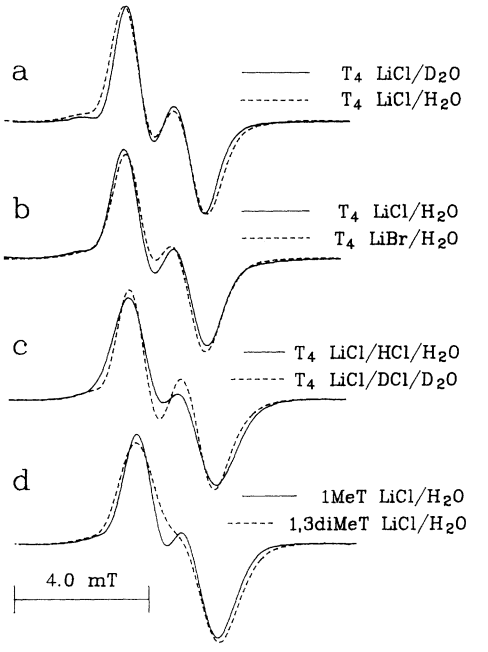


FIG. 2. ESR spectra recorded at 4 K and 35 GHz. a, $d(pT)_4$ in 12 *M* LiCl, H_2O vs D_2O; b, $d(pT)_4$ in 12 *M* LiCl vs 12 *M* LiBr; c, $d(pT)_4$ in 5 *M* LiCl plus 3 *M* HCl/H_2O vs 5 *M* LiCl plus 3 *M* DCl/D_2O; d, 1-MeT in 12 *M* $LiCl/H_2O$ vs 1,3-diMeT in 12 *M* $LiCl/H_2O$.

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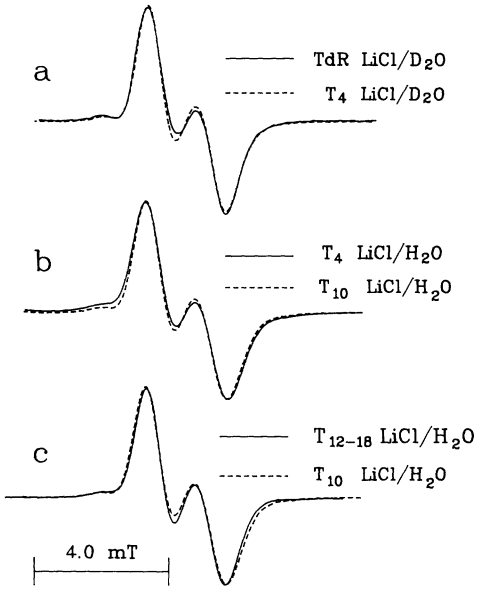


FIG. 3. ESR spectra recorded at 4 K and 35 GHz. All are 12 *M* LiCl glasses. a, TdR vs $d(pT)_4$ using D_2O; b, $d(pT)_4$ vs $d(pT)_{10}$ using H_2O; c, $d(pT)_{10}$ vs $d(pT)_{12-18}$ using H_2O.

marked broadening; and in Fig. 2d, addition of methyl substituents at N1 and N3 gives marked broadening. Variations due to oligomer chain length are shown in Fig. 3 where a slight narrowing (increased doublet resolution) is observed in stepping through the sequence TdR, $d(pT)_4$, $d(pT)_{10}$, and $d(pT)_{12-18}$. The spectra of $d(pT)_2$ and $d(pT)_3$ (not shown) also fit this trend.

Ideally one would like to assign each of these spectral changes to shifts in specific hyperfine or *g* interactions. In practice, this is quite difficult. It is possible, however, to make estimates of the interaction parameters using spectrum simulations combined with information obtained by others on other systems. The hyperfine interactions that play a role are A(H6), A(N1), A(N3), A(H1′), A(H3), A(CH3), and A(HO4).

Simulations. A *Q*-band ESR spectrum of $T^{\bullet-}$ trapped in $d(pT)_4$ in 12 *M* $LiCl/D_2O$ is shown in Fig. 1a. If a simulated spectrum with $R = 0.01$ were superimposed on the experimental spectrum, it would fall within the pen width of the trace. The degree of the fit is better illustrated by plotting the difference between the experimental and the simulated spectra (Fig. 1d). Also shown is the magnitude of the correction due to baseline drift and background (Fig. 1b). The background is broad relative to the $T^{\bullet-}$ spectrum, except for the $D^{\bullet}$ and $H^{\bullet}$ signals. In Fig. 1b the center line of the $D^{\bullet}$ triplet is shown. The outer lines of the $D^{\bullet}$ triplet, and the $H^{\bullet}$ doublet, are used for scaling the background subtraction and serve as a *g*-value reference. The integral of the experimental spectrum, Fig. 1c, illustrates the relatively small degree of baseline distortion.

The simulation, illustrated in Fig. 1, utilized a *g*-tensor, **g**, a linewidth tensor, **W**, an H6 hfc tensor, A(H6), and a nitrogen hfc tensor, A(N). One finds, however, that the elements of A(N) are not very informative, while adding substantially to the computing time. The problem with explicitly incorporating A(N), or any other small hfc

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tensor, into the simulation is that these interactions are not reflected in any distinctive features of the spectrum. To say this another way, there is a large covariance between the elements of small hfc tensors and the elements of the linewidth tensor. It is more efficient and just as informative to incorporate all the weak interaction into **W**. The degree of distortion due to this approximation can be checked by breaking out any of the larger weak interactions. With this strategy we have kept the size of the parameter set minimized. The parameter set consists of the **g,** **A**(H6), and **W** tensors. The R value achieved for $d(pT)_4$ in 12 M $LiCl/H_2O$ is 0.02 with this reduced parameter set. The difference between $R = 0.02$ and $R = 0.01$ for the smaller versus larger parameter set is not considered significant. At this level of fit, larger errors stem from background subtraction, baseline correction, and the existence of multiple minima.

The axes of the simulation have been fixed by forcing the off-diagonal elements of **A**(H6) to zero. The x, y, z axes are defined as shown in Structure I. Since it is known that g_{yy} will be parallel to $A(H6)_{yy}$, it is also possible to set g_{xy} and g_{yz} equal to zero. The **W** tensor is unconstrained.

The simulation parameters and R values for the spectra in Figs. 2 and 3 are given in Table II. The R values, ranging from 0.01 to 0.04, indicate that the fits are very good. The standard deviations given in parentheses reflect the variance of each parameter within the least-squares minimum. The absolute error of these parameters may be significantly larger because of the problem with multiple minima. It is likely, however, that these parameters are reasonable estimates of the actual interactions. More important, variations in the ESR spectra are quantitatively reflected in changes in the simulation parameters, making it possible to use comparisons to estimate weak interactions.

***g** tensor.* If the **g**-tensor parameters are unbounded during the least-squares fit, two different types of **g** tensor are found to yield good fits. In the one case $g_{zz} < g_{yy} < g_{xx}$ while in the other case $g_{yy} < g_{zz} < g_{xx}$. Since it is well established that minimum g, in this type of π-radical, will occur when $\mathbf{B}_0$ is parallel to **y**, the **g** parameters were bounded so as to guarantee that $g_{yy} < g_{zz}$. We emphasize once more that the standard deviations given in Table II are much smaller than the absolute accuracy of the variational parameters. Thus a sense of variability of the **g** elements can be obtained by comparing the results from different samples that should have identical **g,** e.g., T_4 $LiCl/H_2O$ and T_4 $LiCl/D_2O$. The absolute accuracy is thereby no better than ± 0.0002. The Δg's within a tensor, however, are likely to be more accurate. What is needed is a careful measurement of the $T^{\bullet-}$ **g** tensor using single crystal ESR.

A(H6) tensor. Our first attempts to simulate the $T^{\bullet-}$ spectra used the values of **A**(H6) that were determined by ENDOR of TdR (Table I). The simulations, however, gave R values in the 0.10 to 0.20 range. Allowing the three elements of **A**(H6) to vary gave significant improvement with reasonable **A**(H6) values (Table II). Reasonability was judged by comparing the shape of **A**(H6) against the shape expected for π-anion radicals. For example, A_{yy}(H6) is expected to be less than A_{iso}(H6) (*16*). The unpaired electron density at C6, ρ(C6), can be estimated from $A(H6)_{iso}$ (*17*) or, more accurately, from the maximum dipolar coupling ($A(H6)_{iso} - A(H6)_{xx}$) (*16*). For the $T^{\bullet-}$ radical in T_4 LiCl, using **A**(H6) in Table III, one obtains $\rho(C6)_{iso} = 0.52$ and $\rho(C6)_{dip} = 0.53$. A comparison with the values obtained from TdR ENDOR, $\rho(C6)_{iso} = 0.47$ and $\rho(C6)_{dip} = 0.51$, indicates that the spin density at C6 is less sensitive to the N1

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TABLE II

Simulation Parameters for ESR Spectra of T^{-} [a]

	Column Number											
	1	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*	*12*
Solute	T_4	T_4	T_{10}	T_{12-18}	T_4	TdR	T_4	T_4	T_4	T_4	1MeT	1,3diMeT
Matrix	$LiCl/H_2O$	$LiCl/H_2O$	$LiCl/H_2O$	$LiCl/H_2O$	$LiCl/D_2O$	$LiCl/D_2O$	$LiBr/H_2O$	$LiBr/D_2O$	LiCl/HCl	LiCl/DCl	$LiCl/H_2O$	$LiCl/H_2O$
MW band	*Q*	*X*	*Q*	*Q*	*Q*	*Q*	*Q*	*Q*	*Q*	*Q*	*Q*	*Q*
R Value	0.02	0.02	0.02	0.02	0.02	0.04	0.02	0.01	0.02	0.02	0.01	0.01
g_{xx}	2.00339 (5)	2.00404 (6)	2.00342 (5)	2.00346 (2)	2.00351 (2)	2.00359 (3)	2.00346	2.00346 (2)	2.00363	2.00363 (2)	2.00347 (1)	2.00376 (2)
g_{xz}	0.00030 (3)	0.00002 (2)	0.00034 (3)	0.00029 (2)	0.00024 (1)	0.00016 (4)	−0.00011	−0.00011 (1)	0.00016	0.00016 (1)	0.00024 (1)	0.00016 (2)
g_{yy}	2.00241 (6)	2.00198 (6)	2.00226 (5)	2.00250 (1)	2.00225 (2)	2.00216 (4)	2.00234	2.00234 (2)	2.00219	2.00219 (2)	2.00229 (2)	2.00184 (2)
g_{zz}	2.00322 (2)	2.00319 (1)	2.00329 (1)	2.00321 (1)	2.00328 (1)	2.00332 (1)	2.00331	2.00331 (1)	2.00333	2.00333 (1)	2.00333 (1)	2.00350 (1)
A_{xx} (H6)	12.1 (8)	15.5 (2)	14.4 (7)	13.3 (6)	14.2 (2)	12.5 (4)	11.9	11.9 (2)	15.9	15.9 (2)	12.3 (2)	12.0 (2)
A_{yy} (H6)	38.0 (17)	26.7 (2)	38.8 (18)	34.4 (6)	34.6 (4)	32.6 (5)	31.8	31.8 (4)	34.6	34.6 (3)	33.6 (3)	28.7 (6)
A_{zz} (H6)	62.2 (7)	62.6 (1)	57.9 (9)	62.3 (4)	57.5 (3)	59.1 (4)	61.7	61.7 (2)	64.9	64.9 (2)	62.9 (2)	62.3 (3)
W_{xx}	23.2 (2)	21.3 (2)	21.3 (3)	21.5 (4)	20.0 (2)	20.6 (4)	27.7 (2)	23.8 (1)	27.4 (1)	24.1 (2)	24.1 (1)	27.1 (3)
W_{xy}	4.4 (4)	6.8 (2)	1.4 (6)	−0.2 (4)	0.2 (4)	0.1 (8)	−2.8 (4)	0.7 (2)	7.2 (2)	8.6 (4)	4.9 (2)	−4.9 (4)
W_{xz}	−5.4 (2)	−6.7 (1)	−4.8 (3)	−6.9 (4)	−3.9 (2)	−3.1 (6)	8.1 (2)	6.3 (1)	−2.1 (2)	−5.0 (2)	−7.6 (1)	−3.0 (3)
W_{yy}	34.4 (4)	27.7 (2)	32.0 (5)	27.5 (6)	31.7 (4)	32.1 (6)	35.9 (2)	37.4 (2)	46.2 (2)	39.4 (4)	34.4 (2)	38.1 (3)
W_{yz}	−0.7 (5)	−3.1 (3)	0.0 (8)	0.1 (10)	0.0 (6)	0.1 (10)	−1.4 (6)	−5.0 (3)	1.6 (4)	1.4 (3)	1.6 (3)	0.2 (4)
W_{zz}	21.3 (1)	18.3 (1)	20.0 (1)	21.5 (2)	18.5 (1)	18.3 (1)	22.9 (2)	22.2 (1)	25.4 (1)	19.8 (1)	23.5 (1)	25.8 (1)
Diagonalized [b]												
g'_x	2.0036	2.0041	2.0037	2.0037	2.0037	2.0037	2.0035	2.0035	2.0036	2.0036	2.0035	2.0037
g'_y	2.0024	2.0020	2.0023	2.0025	2.0022	2.0022	2.0023	2.0023	2.0021	2.0021	2.0022	2.0017
g'_z	2.0030	2.0032	2.0030	2.0030	2.0031	2.0032	2.0032	2.0032	2.0022	2.0022	2.0031	2.0033
W''_x	26.1	20.1	25.4	28.5	23.2	22.7	28.5	28.5	22.7	25.2	15.1	22.8
W''_y	36.4	34.5	32.2	27.4	31.7	32.1	39.0	39.0	48.6	43.3	36.5	40.1
W''_z	16.4	12.6	15.7	14.6	15.4	16.2	15.8	15.8	27.7	14.9	30.3	28.1
Isotropic												
g_{iso}	2.0030	2.0030	2.0030	2.0030	2.0030	2.0030	2.0030	2.0030	2.0029	2.0029	2.0029	2.0029
A_{iso} (H6)	37.4	34.9	37.0	36.7	35.4	34.7	35.1	35.1	38.5	38.5	36.3	34.3
W_{iso}	26.3	22.4	24.4	23.5	23.4	23.7	28.8	27.8	33.0	27.8	27.3	30.3

[a] Hyperfine couplings are in MHz. Numbers in parentheses are the square root of the variance and correspond to last decimal place; where not given in the upper half of the table those parameters were fixed.

[b] Tensors in the upper half are given in the *A*(H6) principal axis system; the principal values and isotropic components are given below.

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TABLE III

Simulation Parameters for Simultaneous Fit of $d(pT)_4$ and $d(pT)_{10}$ at Q Band and $d(pT)_4$ at X Band[a]

	xx	*xy*	*xz*	*yy*	*yz*	*zz*	*Isotropic*
g	2.0036 (2)	0.0	0.00016 (3)	2.00200 (3)	0.0	2.00335 (1)	2.0030
A (H6)	16.01 (2)	0.0	0.0	35.66 (4)	0.0	57.88 (4)	36.5
W	23.66 (4)	3.84 (5)	−5.03 (4)	30.19 (3)	−2.34 (7)	21.44 (2)	25.1
	Scale	*Freq.*	P_0	P_1	P_2	P_3	P_4
$d(pT)_4$	321.4 (11)	34885.15[b]	0.52×10^1 (27)	0.65×10^1 (4)	-0.31×10^{-3} (5)	-0.74×10^{-4} (4)	-0.27×10^{-6} (1)
$d(dT)_4$	101.7 (8)	9140.89 (6)	0.54×10^1 (35)	-0.32×10^0 (8)	0.26×10^{-2} (8)	0.82×10^{-6} (661)	-0.28×10^{-7} (2)
$d(pT)_{10}$	216.9 (10)	34865.91 (5)	0.71×10^1 (31)	-0.16×10^0 (6)	0.24×10^{-2} (5)	-0.15×10^{-4} (5)	0.31×10^{-7} (16)

[a] Hyperfine couplings are in MHz. Numbers in parentheses are the square root of the variance and correspond to the last decimal place. "Scale" is an amplification factor for the simulated spectrum; "freq." is the microwave frequency in MHz; and P^0–P^4 are the coefficients of a fourth-order polynomial used for baseline correction.

[b] This parameter was not varied.

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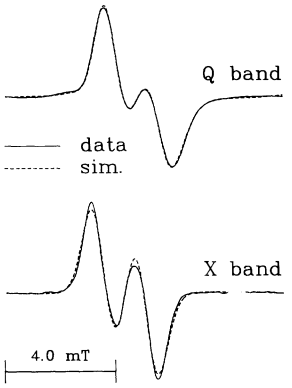


FIG. 4. T_4 in 12 M LiCl/H_2O. ESR spectra taken at Q-band (top) and X-band (bottom) microwave frequencies, T = 4 K. The simulations, done simultaneously, give R = 0.05 (upper) and R = 0.11 (lower) using the parameters given in Table III.

substituent than the spin density adjacent to C6. This is based on the observation that ρ_{iso} is more sensitive to adjacent spin density than is ρ_{dip} (*16*). Since the spin densities at N1, C5, and C6 are sensitive to substituents at N1, they should also be sensitive to rotations about the glycoside bond. Therefore, the torsion angle χ, C2–N1–C1′–04′, is a variable that will affect the ESR characteristics of $T^{\bullet-}$.

W tensor. The linewidth tensor has been used to represent all of the weak hfc's in aggregate. This is an approximation justified largely by being able to achieve low R values using a minimal set of variational parameters. Estimates of various weak (unresolved) hfc's can be made by (1) analyzing changes in **W** obtained through changes in the $T^{\bullet-}$ system, (2) using the dipolar portion of **W**, and (3) drawing on measurements made on aqueous systems. The estimates given below are therefore obtained by a logical unfolding of **W**. These values, not obtained by direct measurement, are given in Table I. The analysis is done using the simulation parameters given in Table III.

The values in Table III were obtained by simultaneously simulating three spectra of the $T^{\bullet-}$ radical: T_4 in LiCl/H_2O at Q band, T_4 in LiCl/H_2O at X band, and T_{10} in LiCl/H_2O at Q band. The goodness of fit is shown in Fig. 4. The R values are 0.05 at Q band and 0.11 at X band. The parameters given in Table III are considered our best estimate, meaning they are likely and reasonable, for simulating the $T^{\bullet-}$ spectrum in a DNA duplex.

A(H3) hfc. The isotropic component of the H3 hfc can be estimated by measuring the change in linewidth of the spectrum of $T^{\bullet-}$ in T_4 LiCl/H_2O versus T_4 LiCl/D_2O. Holding the **g** and **A**(H6) tensors fixed, at the values given in column 5 of Table II, W_{iso} is reduced from 27.4 MHz in H_2O to 23.4 MHz in D_2O. That reduction is due to a collapse in splitting stemming from H3 and neighboring exchangeable protons (H_2O). The change in hyperfine coupling, Δa, is directly proportional to the proton hfc, a_H. (The relation $a_H = 1.4\Delta a$ comes from the two relations $\Delta a = a_H - 2a_D$ and $a_D = 0.15a_H$.) Since Δa = 4.0 MHz, in this case, the total splitting in H_2O that is exchangeable equals 5.6 MHz. A reasonable division of this hfc is $A_{iso}(H3) \simeq 4.0$ MHz and $A_{iso}(H_2O) \simeq 1.6$ MHz.

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A(N1) and A(N3) hfc's. The principal values for nitrogen hfc in a π radical with unpaired electron density on N but not adjacent to N are typically axially symmetric and equal to zero in the nodal plane (A_{yy}(N), 0, 0). That is, the dipolar component B (of $2B$, $-B$, $-B$) equals the isotropic component A_{iso}. If one assumes the anisotropy in **W** is due to nitrogen hfc, then **A**(N) can be estimated. Using the values in Table III, the anisotropy between the plane and its normal is (30.2 − 0.5(23.7 + 21.4)) MHz = 7.7 MHz. If this anisotropy is assigned to the out-of-plane nitrogen hfc and the in-plane hfc is assumed to be zero, then the isotropic nitrogen hfc would be 1.3 MHz. Another way to estimate the isotropic N hfc is by using the values in Table I obtained by aqueous radiolysis. One can use the ratios for change between U and UMP to scale the values measured for T to values expected for TMP. For TMP, by this estimate, one obtains A_{iso}(N1) = 2.5 MHz and A_{iso}(N3) = 0.7 MHz. The total, $A(N)_{iso}$ = 3.2 MHz, is to be compared with 1.3 MHz calculated above. The discrepancy is explainable if a significant portion of the N hfc stems from adjacent spin density, i.e., for N1 the spin density on C6. In-plane N hfc, as observed in closely related radicals (*18*), could easily give a larger N hfc than anticipated from the dipolar component. The estimates in Table III therefore reflect these considerations and utilize the observations made by aqueous radiolysis. We also note that a methyl substituent at N1 and N3 (Table II, columns 11 and 12) appears to affect the N hfc values as well as **A**(H6).

A(H1') hfc. The magnitude of the isotropic H1′ hfc, A_{iso}(H1′), will depend on the C2–N1–C1–H1′ torsion angle, ϕ. To use the approximate equation

$$A_{iso}(H1') = \rho(N1)B_2\cos^2(\phi - 90) \quad (1)$$

one needs an estimate of the N1 spin density, ρ(N1), and the constant B_2 for nitrogen. Using B_2 = 221 MHz (*5*) and calculating ρ(N1) from $A(N1)_{iso} = \rho(N1)Q_N^N$ with Q_N^N = 76 MHz, one obtains $\rho(N1)B_2$ = (0.033)(221) MHz = 7.3 MHz. Therefore, A_{iso}(H1′) should vary from roughly 0 to 7 MHz. Since the torsion angle χ, and therefore the torsion angle ϕ, is likely to vary in oligo(T), one would expect a family of radicals with a range of H1′ hfc's. The χ values for pyrimidine in A-DNA and B-DNA range from −110° to −140° (*19*), putting ϕ between +10° and −20°. The $\cos^2(\phi - 90)$ is therefore small, <0.12, and the H1′ hfc in DNA should be small, <1.0 mHz. We note for example that H1′ hfc was not detected in the ENDOR study of $C^{\bullet-}$ in 5′dCMP (*20*) or in the $U^{\bullet-}$ anion in aqueous 5′UMP (*10*). The value of A_{iso}(H1′) = 2.0 MHz given in Table I is an estimate based on the value falling in the low part of the 0- to 7-MHz range.

A(CH_3) hfc. Although the LiCl glass data cannot be used to calculate $A_{iso}(CH_3)$ directly, it can be used to obtain upper and lower limits. (If budget permitted, deuteration of the $T^{\bullet-}$ methyl would provide a means of directly measuring $A_{iso}(CH_3)$.) Our best estimate is obtained by subtracting each of the hfc values, discussed above, from W_{iso}. This calculation is laid out in Table IV along with the two extremes. The conclusion is that $A_{iso}(CH_3)$ falls between 1.7 and 6.3 MHz with 3.7 MHz as our best estimate.

pH effect. High pH glasses, 3 *M* HCl + 5 *M* LiCl, give clear evidence of protonation of $T^{\bullet-}$ at O_4. From the change in linewidth (Table II, columns 9 and 10), A_{iso}(H4′) can be estimated to be ~7 MHz (from 1.4 × (33.0–27.8 MHz)). A_{iso}(H6) increases 1–3 MHz upon 04 protonation.

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TABLE IV

Range of Weak Hyperfine Couplings in $T^{\bullet-}$

Hyperfine coupling	*Hyperfine coupling multiple*	*Best estimate*	*Maximized* $A_{iso}(CH_3)$	*Minimized* $A_{iso}(CH_3)$
Matrix	1	1.6	1.0	2.0
A_{iso}(H3)	1	4.0	1.0	4.0
A_{iso}(H1′)	1	2.0	0.0	6.0
A_{iso}(N1)	2	2.5	1.6	3.0
A_{iso}(N3)	2	0.7	0.5	1.0
$A_{iso}(CH_3)$	3	3.7	6.3	1.7
Total		25.1	25.1	25.1

Substituent effects. The broadening effects of adding a methyl substituent to N1 and N1 plus N3 are shown in Fig. 2d. The simulation given in Table II indicates this is partly due to an increase in A_{dip}(H6) and partly due to an increase in **W**. Therefore, it is likely that methyl substitution is acting as was observed in uracil (*10*), increasing the hfc to N1 and N3, decreasing ρ(C5), increasing ρ(C6), and adding very small CH_3 hfc from the N methyls. The LiCl glass data show that changes of this magnitude do not occur when $T^{\bullet-}$ is trapped in a polymer versus a monomer environment.

Conformation effects. There is a small but measurable increase in doublet resolution of the $T^{\bullet-}$ spectrum as oligo(T) is increased in length. In going from $d(pT)_4$ to $d(pT)_{12-18}$, W_{iso} decreases by 2.8 MHz and A_{iso}(H6) decreases by 0.7 MHz. We suggest that this is primarily due to a narrowing in the range of conformations, the main variable being rotation of the base about the glycosidic bond (torsion angle χ). The suggestion is that χ rotations can account for shifts in the weak hfc that in aggregate might total ~3 MHz and a shift in H6 hfc of 1–5 MHz. (For A_{iso}(H6) we compare the TdR ENDOR value of 32.8 MHz with those in Tables II and III.) The slight redistribution of unpaired electron density due to χ rotation can be compared with substituent effects. Rotating through 360° is not likely to change the spin distribution as much as methyl substitution at N1, for example.

It appears unlikely that further lengthening of the oligomer chain will lead to significant changes. In Fig. 5 the spectrum of poly(T) in LiCl is compared with $d(pT)_{10}$. Although the background in poly(T) is much greater than $d(pT)_{10}$, the $T^{\bullet-}$ spectrum appears to change very little if at all.

Matrix effect. Glasses made from LiBr give a broader $T^{\bullet-}$ spectrum than glasses made from LiCl. In the case of T_4, W_{iso} increases by 2.5–4.4 MHz (in Table II compare columns 1 and 7 and columns 5 and 8). This effect could be due in part to conformational changes and in part to direct interaction between the halogen and $T^{\bullet-}$ radical. To view $T^{\bullet-}$ in the absence of LiCl, we irradiated powdered poly(T) pressed into a pellet. Although there is now a background due to thymine radical cations, it is evident from Fig. 5c that no major changes in $T^{\bullet-}$ occur.

DISCUSSION

Through the simulation of randomly oriented $T^{\bullet-}$ radicals, it is possible to estimate the various interactions that comprise the ESR spectra. These estimates depend

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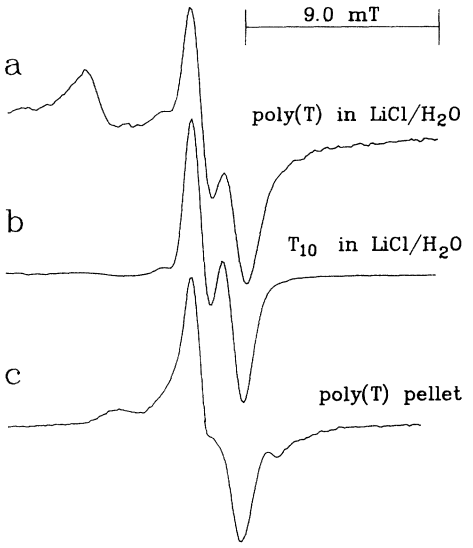


FIG. 5. ESR spectra taken at 4 K and 35 GHz: a, Poly(T) in 12 *M* LiCl; b, $d(pT)_{10}$ in 12 *M* LiCl; and c, pellet of compressed poly(T) powder.

heavily on information from three sources: (1) the spectral variations brought about by known chemical changes, (2) extrapolation from aqueous radiolysis and ENDOR measurements, and (3) general knowledge of relevant hfc and **g** tensors. The estimates given in Table I, row 8, are therefore not as accurate as implied by the number of decimal places given. This should be apparent from the range of values given in Table IV. The numbers given in Table I, however, are self consistent with the LiCl data, aqueous radiolysis data, and our current knowledge on these types of interactions. What is apparent is that the values stemming from our analysis deviate significantly from those determined from the DNA fiber analyses.

The strategy behind using DNA packaged in a bundle of fibers is to enhance the spectral information by aligning the DNA so that the bases are coplanar. With all the bases coplanar, the spectrum taken with B_0 perpendicular to the base plane will be of single crystal quality and the spectrum with B_0 parallel to the base plane will be due to two-dimensional instead of three-dimensional tensors. In actuality the bases are not entirely coplanar. It is necessary, in A-DNA, to introduce parameters to account for base tilt relative to the helix axis and, in all cases, to account for regions where DNA itself is not parallel to the fiber package axis (*12*). Thus the advantage of enhanced spectral structure is partially lost in the necessity of increasing the number of variational parameters. In addition, the $T^{\bullet-}$ spectrum is always superimposed on the spectrum of one or more other radicals. The task of simulating a composite spectrum, consisting of two or more superimposed spectra, is very difficult. Analysis of the DNA data would be facilitated if the spectral parameters of the $T^{\bullet-}$ radical were known.

A comparison of the simulation parameters determined from the DNA fiber data with those from the LiCl glass data shows the largest discrepancy is in $A_{iso}(CH_3)$. The point that the CH_3 hfc of 10–11 MHz is almost certainly too large has been made before (*5*), but this is not the key issue. Simulation of the DNA fiber data is a multi-

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minimum problem. This is apparent just from the two solutions shown in Table I. If one lets the **g** tensor, linewidth tensor, and two or three hfc tensors vary along with base alignment parameters, there will be multiple minima. If the small hfc parameters are not taken literally and are instead rolled into a tensor **W**, then the discrepancies between the fiber and glass simulations are reduced. We used the DNA fiber parameters given in Table I, row 7, to simulate the *X*- and *Q*-band spectra shown in Fig. 4. The linewidth tensor **W** was allowed to vary in order to optimize the fit. The *R* values thus obtained are 0.30 for *X* band and 0.15 for *Q* band, indicating that the major features of the $T^{\bullet-}$ spectra are simulated. The fit improves if variation of **g** and **A**(H6) is permitted. Our conclusion is that the simulation parameters determined from fiber data, although perhaps incorrect with regard to specific small hfc's, do a reasonable job in simulating the primary features of the $T^{\bullet-}$ spectrum, and in doing so provide evidence for the presence of $T^{\bullet-}$ in DNA fibers at 77 K.

If the parameters presented in Table III are to be useful in quantifying the $T^{\bullet-}$ contribution in DNA, then it is important to determine to what degree, if any, the parameters change when $T^{\bullet-}$ is located in DNA. Our analysis of the oligo(T) and poly(T) data indicates that variations due to conformation and matrix are not large enough to account for the differences in simulation parameters found for fiber DNA versus LiCl glasses. An experiment that would shed some light on this would be to simulate the DNA fiber data using the parameters in Table III. This, in conjunction with a simulation of a spectrum of the same DNA but as a random powder instead of as a fiber bundle, should help to resolve these discrepancies. Of particular interest is the fiber simulation with B_0 normal to the base planes. At this orientation the fiber parameters predict a hyperfine pattern composed of a 36-MHz doublet (1:1), a 12-MHz triplet (1:1:1), an 11-MHz quartet (1:3:3:1), and a 9-MHz doublet (1:1). The pattern is approximately 10 lines with intensity ratios of 1:5:11:15:16:16:15:11:5:1 and a spectral width of 112 MHz. The splitting is large enough that this hyperfine pattern must be simulating distinctive features of the DNA fiber spectrum. If the $A_{iso}(CH_3)$ is too large, then these features must, at least in part, be due to other radical products. Two possibilities, for example, are the N6 deprotonated cytosine cation and the N2 deprotonated guanine cation.

Another important aspect of electron trapping by thymine in DNA is the protonation state of the electron adduct. Since there is evidence that $T^{\bullet-}$ will readily protonate at O4, it is important to be sure whether or not protonation at O4 can be discerned with ESR. Differences in the spectra of thymine electron adducts under basic, neutral, and acidic conditions have been noted (*8*). Since protonation at O4 causes measurable changes in the oligomer system, we conclude that, at neutral pH, the major fraction of $T^{\bullet-}$ trapped in oligo(T) or poly(T) is not protonated at O4 in the 4 to 77K temperature region. This observation is important to determining the factors which govern the trapping sites. It means that, at temperatures below 78 K, protonation does not play a key role in electron trapping at the thymine site and that $T^{\bullet-}$ probably retains its negative charge until protonation occurs at C6, as originally proposed (*1*). Also, from the perspective of the opposing adenine base, in duplex DNA, the adenine base remains neutral. This could be important because proton transfer to O4 of thymine would result in a negative charge on the opposing adenine. That negative charge will influence the distribution of the electron and hole trapping in its immediate vicinity.

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The results presented here redirect attention to the possibility that thymine is not the only site of electron trapping at temperatures below 78 K. In the early DNA fiber work (*11*), cytosine was maintained to be a possible trapping site along with thymine. Sevilla *et al.* noted that $C^{\bullet-}$ could not be readily distinguished from $T^{\bullet-}$ in their studies of one-electron reduced dinucleoside phosphates (*4*). As we have stressed, both $C^{\bullet-}$ and $T^{\bullet-}$ are well characterized by single crystal ESR, and there is no evidence as of yet that would preclude cytosine as a site of electron trapping in DNA (*5*). We are now in the process of studying one-electron reduced cytosine and oligomers of mixed base composition. The preliminary findings are that at <78 K cytosine, not thymine, is the dominant site of electron trapping (*21*). This correlates with recent measurements showing that cytosine has a slightly larger reduction potential than thymine.[1] It is feasible that electron trapping at low temperatures occurs at all the bases, with a distribution governed by the reduction potentials and relative proton donor-acceptor properties of the DNA base pairs.[1]

CONCLUSIONS

The thymine radical anion, $T^{\bullet-}$, does not protonate in oligo(T) or poly(T) at temperatures between 4 and 77 K. Changes in conformation and trapping environment produce relatively minor changes in the $T^{\bullet-}$ ESR spectrum. In the LiCl glasses, and by inference in the previously published DNA fiber system, it does not appear possible, using existing ESR data, to specify a unique set of hyperfine and *g* interactions. It is possible, however, to determine boundaries for these interactions. A set of simulation parameters that are likely and reasonable are presented that can be used to simulate $T^{\bullet-}$ ESR spectra in samples containing randomly oriented radicals.

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